

Project Title -

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PROBLEM STATEMENT

- (Agricultural activities are influenced by seasonal variations in environmental conditions, farming practices, resource availability and market conditions. As a result, agricultural performance may differ from one season to another. However, raw agricultural data does not clearly explain how agricultural performance changes across seasons or what patterns can be observed in different seasonal conditions. The problem is to analyze the given agricultural dataset and investigate seasonal differences in agricultural performance by identifying meaningful patterns, trends, relationships and variations within the available data.



Project Description:-

Analyzes agricultural performance across different seasons.

Examines the impact of environmental conditions, farming practices, and resource usage.

Compares crop yield, production, water efficiency, revenue, and profit across seasons.

Uses data visualization and statistical analysis to identify patterns and relationships.

Provides data-driven insights and recommendations for better seasonal agricultural planning.

WHO ARE THE END USERS?



-  Farmers - For better crop and resource planning.
-  Agricultural Departments - For seasonal agricultural planning and policy decisions.
-  Agricultural Researchers - To study seasonal patterns and crop performance.
-  Resource Managers - To improve water, fertilizer, and pesticide usage.
-  Agribusinesses - For production and market planning.



Technology Used

- Python - Data analysis and programming
- Google Colab - Development environment
- Pandas & NumPy - Data processing
- Matplotlib & Seaborn - Data visualization
- SciPy - Statistical analysis
- Excel/CSV - Dataset format



RESULTS

Identified seasonal variations in crop yield and production.

Compared rainfall, temperature, humidity, and soil conditions across seasons.

Analyzed fertilizer, pesticide, nutrient, irrigation, and water usage.

Evaluated water-use efficiency to identify efficient farming practices.

Compared revenue, cost, market price, and profit across seasons.

Identified the relationship between environmental factors and crop yield.

Analyzed disease and pest risk across different seasons.

Used statistical and correlation analysis to identify significant patterns.

Identified better-performing and lower-performing seasons.

Generated data-driven recommendations for improving agricultural productivity and profitability.

RESULTS

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

```
df = pd.read_csv("/content/seasonal_agriculture_performance_dataset.csv")
```

```
df.head()
```

	Farm_ID	State	District	Crop	Season	Farm_Area_Hectares	Rainfall_mm	Avg_Temperature_C	Humidity_pct	Sunlight_Hours_Day	...	Seed_Quality_Score	Yield_Tonnes_Ha	Production_Tonnes	Market_Premium
0	SF10001	Andhra Pradesh	Rajkot	Wheat	Kharif	0.53	486.4	24.3	69.1	5.8	...	0.76	2.46	1.30	0.15
1	SF10002	Maharashtra	Nalgonda	Maize	Kharif	6.53	855.4	25.7	78.4	5.1	...	0.94	0.30	1.96	0.20
2	SF10003	Telangana	Warangal	Pulses	Rabi	4.86	455.6	23.9	56.4	10.3	...	0.98	0.52	2.53	0.10
3	SF10004	Telangana	Indore	Rice	Kharif	4.50	753.2	29.7	65.2	6.3	...	0.74	1.84	8.28	0.25
4	SF10005	Karnataka	Ludhiana	Maize	Zaid	4.21	101.6	34.1	55.2	6.5	...	0.90	2.21	9.30	0.18

5 rows × 28 columns

```
df.tail()
```

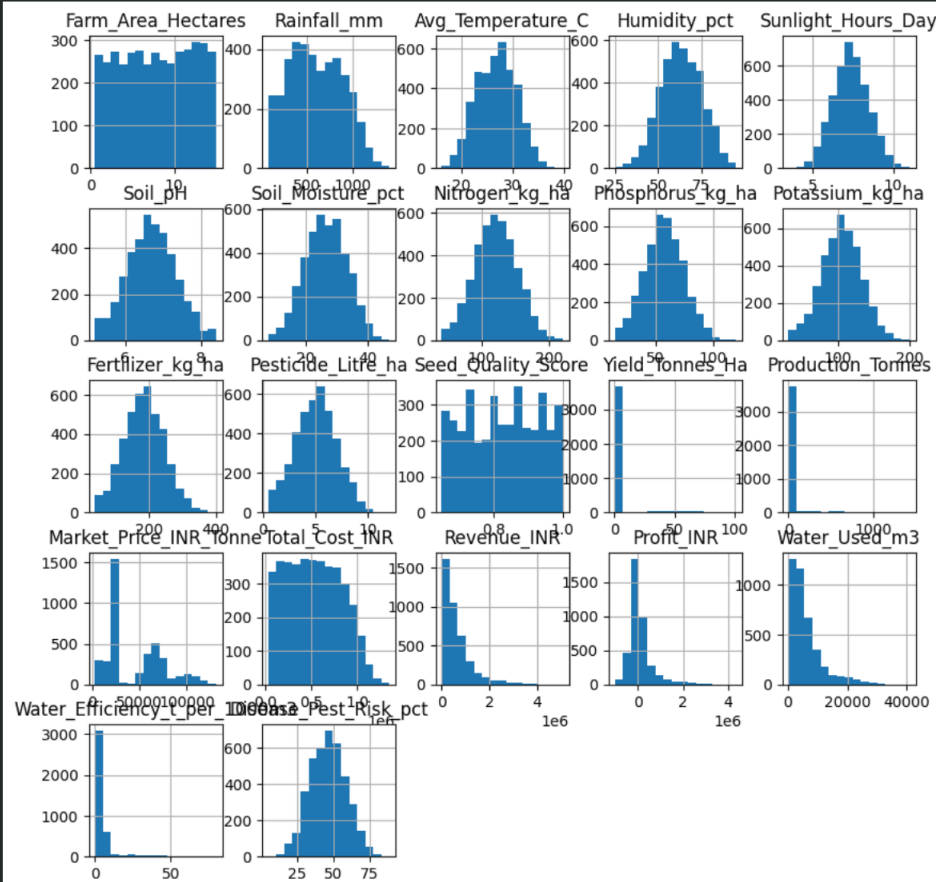
...	Farm_ID	State	District	Crop	Season	Farm_Area_Hectares	Rainfall_mm	Avg_Temperature_C	Humidity_pct	Sunlight_Hours_Day	...	Seed_Quality_Score	Yield_Tonnes_Ha	Production_Tonnes	Market_Premium
3995	SF13996	Karnataka	Rajkot	Wheat	Kharif	4.72	865.4	27.3	51.4	6.3	...	0.74	0.51	2.41	0.12
3996	SF13997	Karnataka	Krishna	Chilli	Kharif	8.77	947.5	29.2	85.5	6.1	...	0.80	2.75	24.12	0.22
3997	SF13998	Andhra Pradesh	Erode	Pulses	Zaid	10.47	80.0	35.4	68.5	8.6	...	0.99	0.62	6.49	0.18
3998	SF13999	Punjab	Warangal	Cotton	Kharif	1.73	475.0	28.2	61.7	6.4	...	0.98	1.35	2.34	0.15
3999	SF14000	Punjab	Nashik	Wheat	Kharif	3.31	984.3	25.8	77.2	5.9	...	0.99	2.36	7.81	0.20

5 rows × 28 columns

RESULTS

```
df.hist(bins=15,figsize=(10,10))
```

```
<Axes: title={'center': 'Yield_Tonnes_Ha'}>,  
<Axes: title={'center': 'Production_Tonnes'}>],  
[<Axes: title={'center': 'Market_Price_INR_Tonne'}>,  
<Axes: title={'center': 'Total_Cost_INR'}>,  
<Axes: title={'center': 'Revenue_INR'}>,  
<Axes: title={'center': 'Profit_INR'}>,  
<Axes: title={'center': 'Water_Used_m3'}>],  
[<Axes: title={'center': 'Water_Efficiency_t_per_1000m3'}>,  
<Axes: title={'center': 'Disease_Pest_Risk_pct'}>],  
<Axes: >], dtype=object)
```

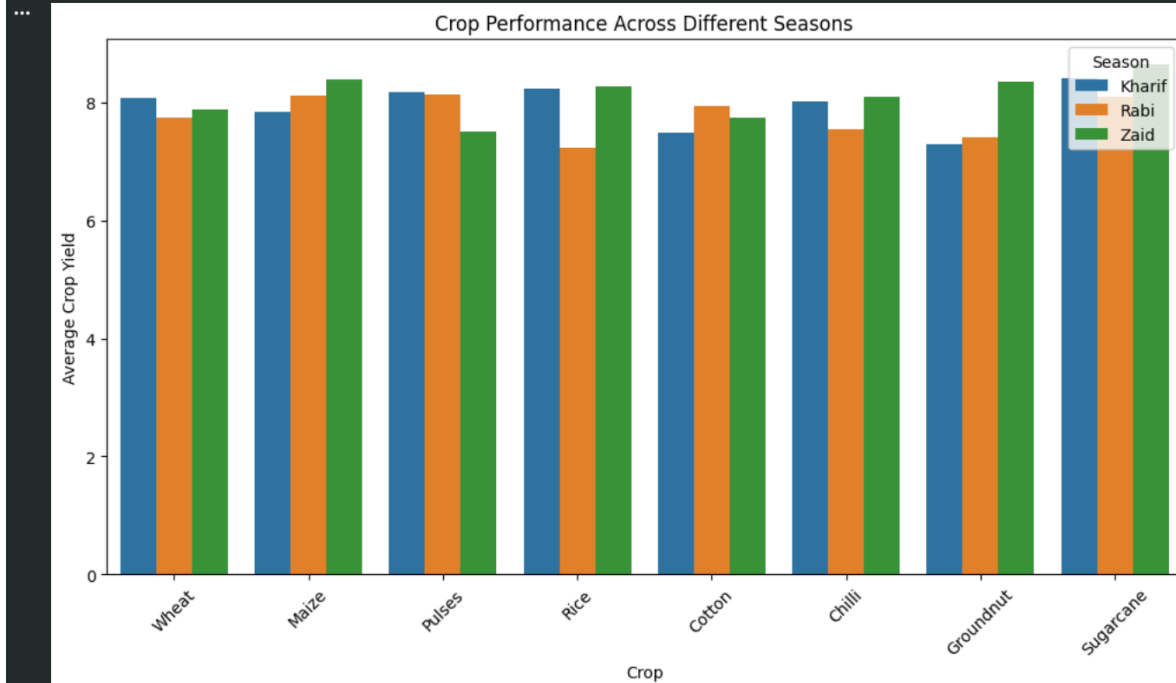


RESULTS

```
plt.figure(figsize=(12,6))
```

```
sns.barplot(  
    data=df,  
    x='Crop',  
    y='Farm_Area_Hectares',  
    hue='Season',  
    errorbar=None  
)
```

```
plt.title('Crop Performance Across Different Seasons')  
plt.xlabel('Crop')  
plt.ylabel('Average Crop Yield')  
plt.xticks(rotation=45)  
  
plt.show()
```



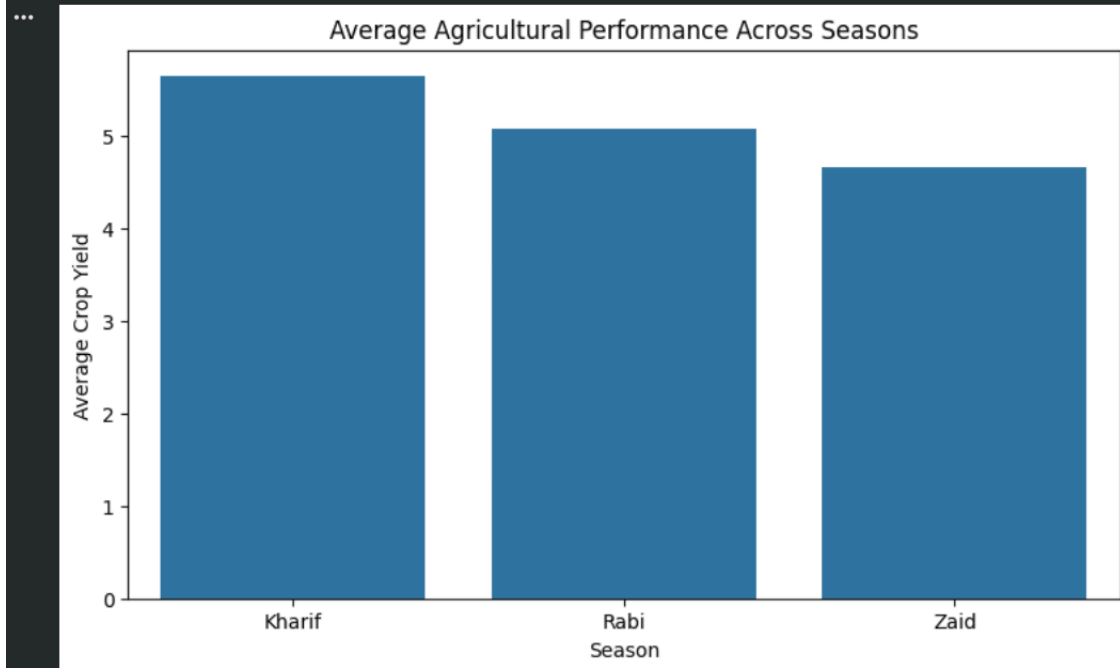
RESULTS

```
plt.figure(figsize=(9,5))

sns.barplot(
    data=df,
    x='Season',
    y='Yield_Tonnes_Ha',
    estimator='mean',
    errorbar=None
)

plt.title('Average Agricultural Performance Across Seasons')
plt.xlabel('Season')
plt.ylabel('Average Crop Yield')

plt.show()
```



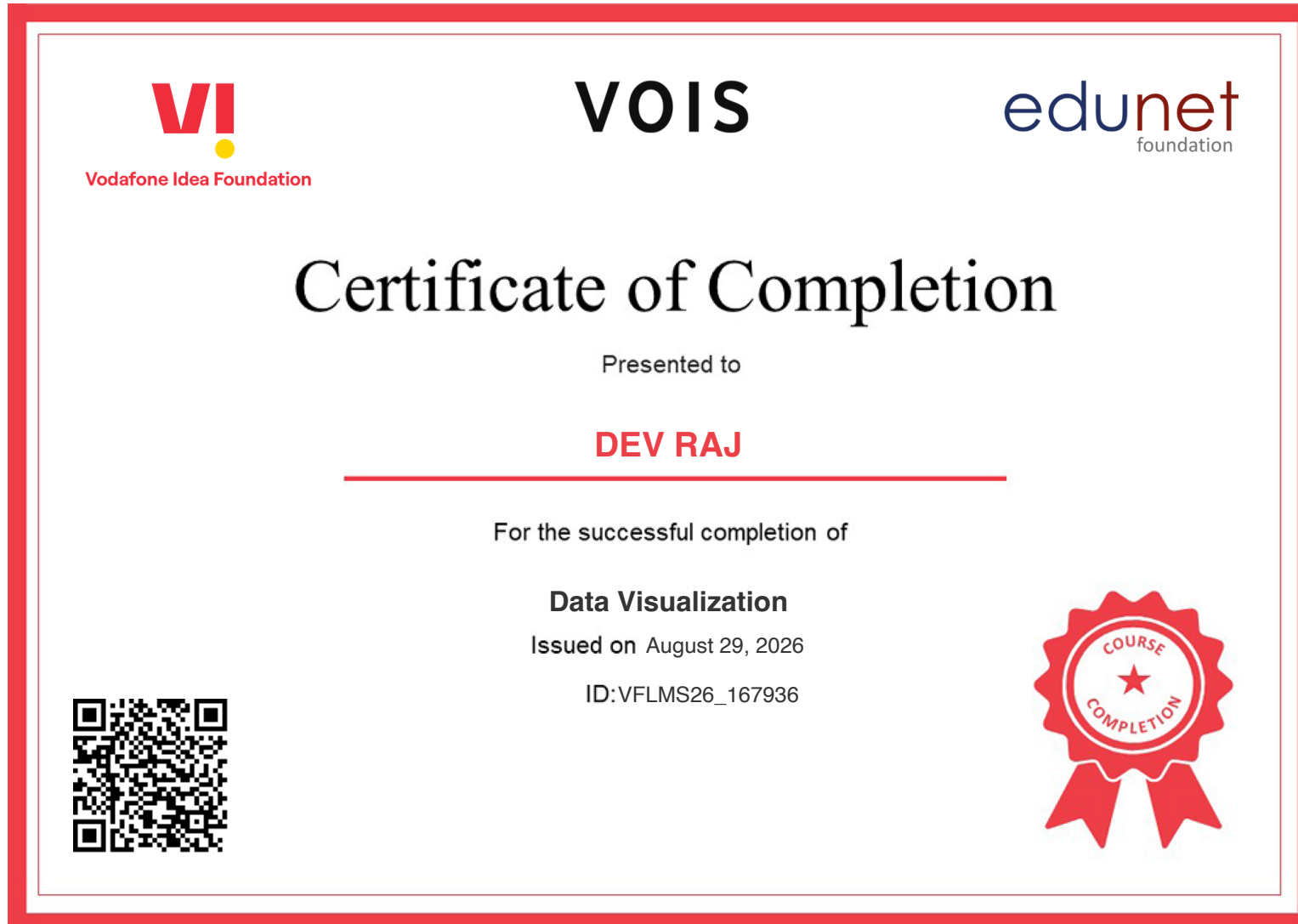
Future scope

-  Analyze crop performance across seasons 
-  Study environmental conditions and their impact
-  Evaluate water and resource efficiency
-  Compare cost, revenue, and profitability
-  Analyze disease and pest risks
-  Identify trends, patterns, and relationships
-  Provide data-driven recommendations for better farming decisions

GitHub Link

https://github.com/devraj93341/Major-Project_Seasonal-Agriculture-Performance-Analysis

VOIS Course completion certificate - Course name- Data Visualization





Thank you

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